

FJM30 Wideband Integrated RF Processing Device

Complete Product & Sales Brochure | Alight Semiconductor

1. What is FJM30?

FJM30 is a highly integrated wideband RF processing device that combines:

- **Wideband RF transceiver** (50MHz – 6GHz)
- **FPGA programmable logic**
- **ARM processing system**
- **High-speed DSP capability**

...into a single compact device.

In simple terms, FJM30 replaces the traditional multi-chip SDR architecture:

RF Transceiver + FPGA + ARM Processor + DDR + Peripheral Control

...with **one highly integrated solution**.

You can think of FJM30 as an **AD9361 and a Zynq combined into one chip** – designed for engineers who want to build compact, low-power SDR platforms without struggling with multi-chip integration.

2. Company Background

Alight Semiconductor was founded in **2016**, specializing in **System-in-Package (SiP)** technology.

For the first several years, the majority of our customers came from **specialized sectors in China** where public disclosures are not permitted. This is why you will not find many public case studies or Western media coverage.

Despite the lack of public visibility, our SiP technology has been **mass-produced and field-proven** in demanding environments for years.

✓ We are not a new startup – we are an experienced SiP specialist now opening our technology to global commercial markets.

3. Strategic Shift to Global Markets (2024)

Starting in **2024**, Alight Semiconductor made a strategic decision to **commercialize our technology globally**.

FJM30 is our flagship product for this new direction.

Since then:

- Multiple engineering teams **outside of China** have completed validation
- Several customers have moved to **volume production**
- They chose FJM30 because it **significantly reduces development time and BOM complexity** – benefits that are immediately measurable

We are now actively engaging with global SDR, wireless infrastructure, and defense/commercial radio developers.

4. Key Advantages of FJM30

Area	Benefit
System Size	Much smaller form factor – ideal for compact SDR and embedded wireless systems
Integration Complexity	Reduces high-speed interconnect and mixed-signal design difficulty
Development Cycle	Faster – teams focus on algorithms and applications, not board-level integration
System Cost	Lower BOM, fewer PCB layers, simpler power circuitry and assembly
Power Consumption	Lower – suitable for portable, edge, and power-sensitive applications
SDR Flexibility	Supports FPGA-based custom waveform, DDC/DUC, spectrum analysis, protocol acceleration

5. What Traditional Architectures Can FJM30 Replace?

FJM30 is designed to replace:

- AD9361 + FPGA solutions
- Discrete SDR architectures
- Large RFSoc-based systems where cost/power are excessive
- Multi-board RF processing platforms
- Custom wireless communication systems requiring compact integration

✓ Instead of two separate chips connected via LVDS, FJM30 integrates the RF transceiver, dual-core ARM, and 85K logic cells into a single **17×17mm BGA**.

6. Technical Specifications Summary

Heterogeneous Processing Architecture

Component	Specification
ARM Subsystem	Dual-core ARM, up to 866 MHz, 32-bit, L1 cache, AXI interconnect
Programmable Logic	85K equivalent logic cells, 200 hardware multipliers
Memory	External DDR3/DDR3L/LPDDR2 (16-bit, up to 1066 MT/s), 5Mb on-chip RAM, 256Mb SPI flash

RF Subsystem (2T2R Direct Conversion)

Parameter	Value
Frequency range	50 MHz – 6.0 GHz
Channel bandwidth	200 kHz – 56 MHz
Aggregate usable BW	100 MHz
ADC/DAC resolution	12-bit
RX Noise Figure	2.4 dB @1.5 GHz, <5.6 dB across band
TX max output power	+5 dBm (typical)
TX EVM	<1% @1.5 GHz, <2% across band
Frequency resolution	2.4 Hz
Phase noise	-108 dBc/Hz @10kHz offset (1.5 GHz)
Hopping rate	up to 10,000 hops/s
TDD/FDD support	Yes

Digital Signal Processing Support

- DDC / DUC
- FIR/CIC filtering
- FFT-based spectral analysis
- Custom protocol acceleration
- Waveform generation (AM/FM, PSK/QAM, custom)
- Digital IQ imbalance and DC offset correction

Interfaces & Power

- **Interfaces:** Gigabit Ethernet, USB 2.0 OTG, SD/SDIO, CAN, UART, SPI, I²C, GPIO
- **Power Supply:** 1V (1.5A max), 1.8V (2A max), 3.3V (1A max)
- **Package:** 625-ball BGA, 17×17×1.66 mm, 0.65mm pitch
- **Software Support:** Embedded Linux, RTOS, bare-metal
- **Development Flow:** Standard FPGA toolchains (Verilog/VHDL), JTAG debug, integrated logic analysis, remote firmware update

7. Comparison with Traditional Architecture

Aspect	Traditional (AD9361 + Zynq)	FJM30 Single-Chip
PCB size	Large	Very small (17X17mm)
LVDS routing	Required	None
Signal integrity debug	Complex	Eliminated
Software environment	Linux/Petalinux	Same – no rewrite
External memory	Mandatory	Optional (DDR controller integrated)
BOM complexity	High (multiple vendors)	Low (one chip)
Hardware team dependency	RF + FPGA specialists	Single team

8. Evaluation Board & Sales Strategy

Risk Mitigation for Customers

We fully understand that adopting a new chip – especially from a less-publicized company – involves **risk concerns**.

To address this, we provide **official evaluation boards (EVKs)** for customers to perform:

- Functional testing
- Performance verification
- Software and algorithm development
- Integration trials

Evaluation Board Payment Terms

The payment terms for the evaluation board are designed to be **flexible** and **should not create a significant burden** for an engineering team or project budget.

We are open to discussing arrangements that work for your procurement process.

What You Get with the EVK

- FJM30-based evaluation board
- Power supply and basic I/O access
- Linux/Petalinux BSP
- JTAG debug interface
- Documentation and quick-start guide
- Remote firmware update support

Next Step:

If you are interested, we can send you the official evaluation board purchase proposal for your reference. This includes pricing, delivery schedule, and technical support terms.

9. Addressing Common Concerns

Concern	Response
No public case studies?	Our prior customers were in restricted sectors in China. We are now opening to global markets with FJM30.
Is this a new, untested platform?	No – it is based on the traditional RF+FPGA+ARM architecture you already know. The methodology is the same.
What if it doesn't work for us?	Start with the evaluation board. Low risk, flexible terms.
How is supply chain risk?	Single vendor, single chip. Simpler than managing ADI + Xilinx + memory vendors.

10. Applications

- MIMO multi-polarization transceiver communication
- Software-defined radio (SDR) platforms
- Multi-channel digital/analog signal sources
- Real-time spectrum analyzers and frequency counters
- Custom wireless communication systems
- Portable and edge RF systems

11. Block Diagram & Package Information

Block diagram reference: See Figure 1 in product datasheet.

- **Package Type:** BGA
- **Dimensions:** 17.0 × 17.0 × 1.66 mm
- **Ball Count:** 625 balls
- **Pitch:** 0.65mm

12. Summary - Why FJM30?

Traditional Approach	FJM30 Single-Chip Solution
3+ chips from 2+ vendors	1 chip, 1 vendor
Complex LVDS routing	No LVDS
Large PCB	17×17mm BGA
High power	Optimized low power
Long debug cycle	Fast integration
Public data available	EVK available for validation
Well-known	New but proven in production

13. Contact & Next Steps

Alight Semiconductor

Founded 2016 | SiP Specialist | Now Global

- ✓ Evaluation boards available
- ✓ Flexible payment terms for EVKs
- ✓ Linux/Petalinux ready
- ✓ Volume production support
- ✓ Direct engineering support available

✉ Request evaluation board purchase proposal →

📄 Request technical datasheet →

☎ Request engineering call →

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